

1. user model :

familyName: String
firstName: String
email: String
password: String
birthday: Date
sexe: String (enum: 'Homme' | 'Femme')
profileImage: String
userName: String
role: String (enum: 'admin' | 'user', default: 'user')
verified: Boolean (default: false)

2. article model :

title: String
description: String
scholarName: String
epoque: String (enum of historical periods)
language: String (enum: 'Arabe' | 'Français' | 'Tamazight' | 'Bilingue' | 'Multilingue')
domaineExpertise: String (enum of disciplines)
imageArticle: {
 url: String
}
sections: Array of sectionSchema
pdfFiles: Array of {
 url: String
}
author: ObjectId (ref: 'User')
status: String (enum: 'draft' | 'pending' | 'approved' | 'rejected' | 'archived')
adminReview: {
 reviewedBy: ObjectId (ref: 'User')
 reviewedAt: Date
 reviewNotes: String
 rejectionReason: String
}
publishedAt: Date (default: null)

```
viewCount: Number (default: 0)
likes: Array of {
  user: ObjectId (ref: 'User')
  likedAt: Date (default: Date.now)
}
comments: Array of commentSchema
featured: Boolean (default: false)
readingTime: Number (in minutes)
viewers: Array of {
  user: ObjectId (ref: 'User')
  viewedAt: Date (default: Date.now)
  ip: String
  userAgent: String
}
```

2.1. section model :

```
title: String
content: String
pictures: Array of {
  url: String
}
videos: Array of {
  url: String
}
order: Number
```

2.2. comment model :

```
content: String
author: ObjectId (ref: 'User')
parentComment: ObjectId (ref: 'Comment', default: null)
likes: Array of {
  user: ObjectId (ref: 'User')
  likedAt: Date (default: Date.now)
}
isEdited: Boolean (default: false)
editedAt: Date
status: String (enum: 'active' | 'hidden' | 'deleted', default: 'active')
```